4/17/2026

Bash Scripting Project

User and Group Management Toolkit

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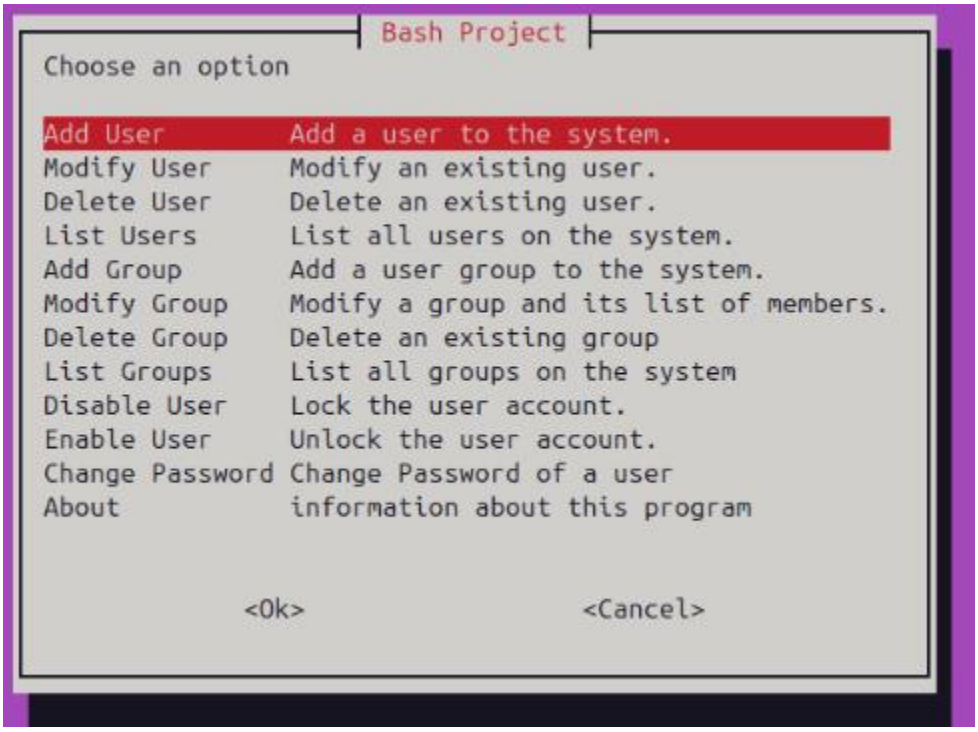
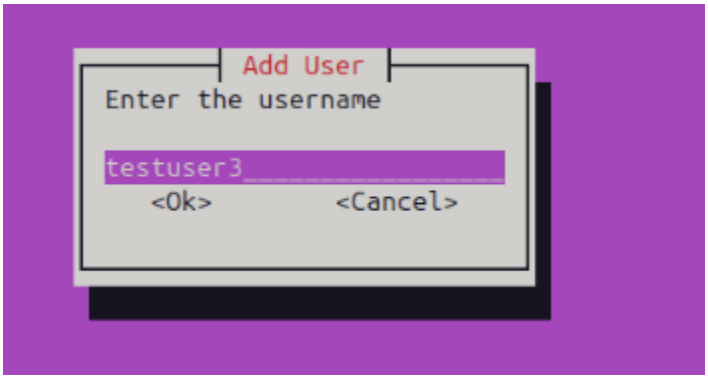
SYSTEM ADMINISTARTATION

1- User interface:




```bash
#!/bin/bash

OPTION=$(whiptail --title "Bash Project" --menu "Choose an option" 20 60 12 \
"Add User" "Add a user to the system." \
"Modify User" "Modify an existing user." \
"Delete User" "Delete an existing user." \
"List Users" "List all users on the system." \
"Add Group" "Add a user group to the system." \
"Modify Group" "Modify a group and its list of members." \
"Delete Group" "Delete an existing group" \
"List Groups" "List all groups on the system" \
"Disable User" "Lock the user account." \
"Enable User" "Unlock the user account." \
"Change Password" "Change Password of a user" \
"About" "information about this program" \
3>&1 1>&2 2>&3)

echo "You chose: $OPTION"
```

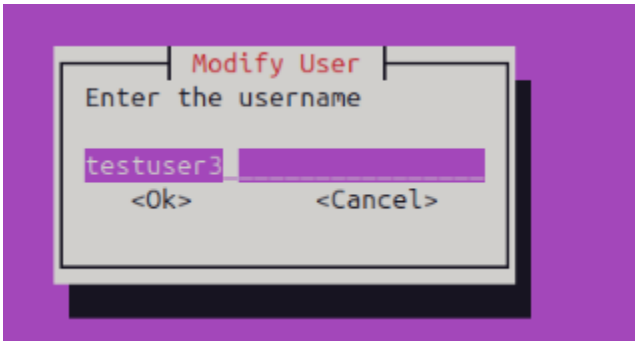
2- Add User

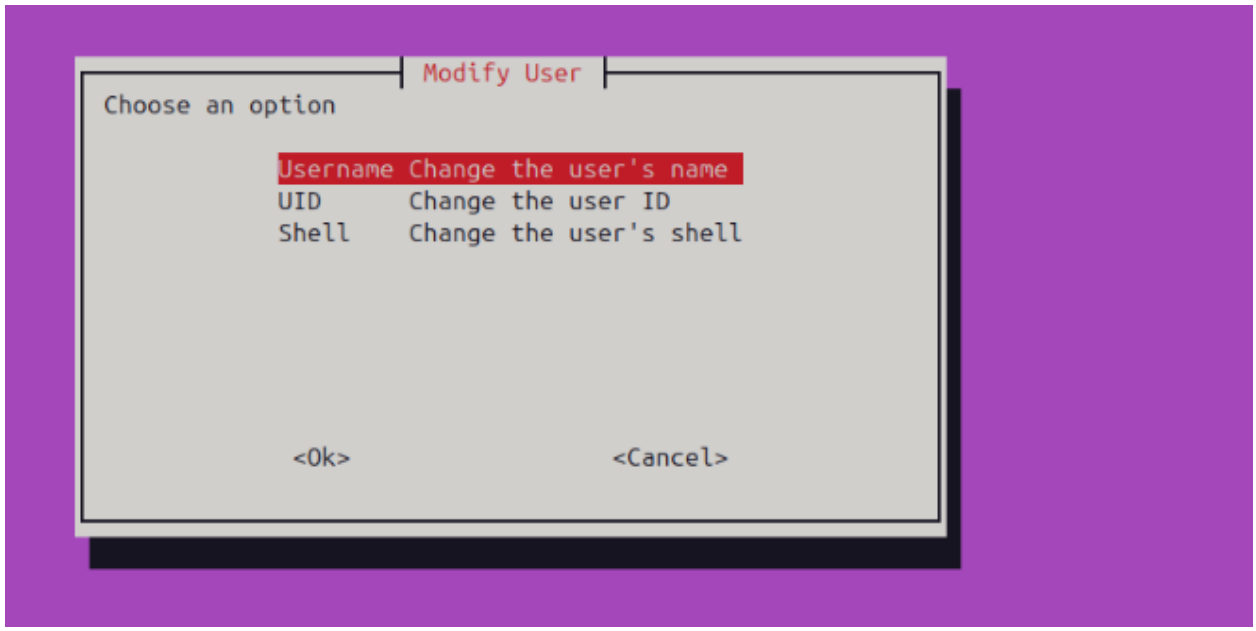


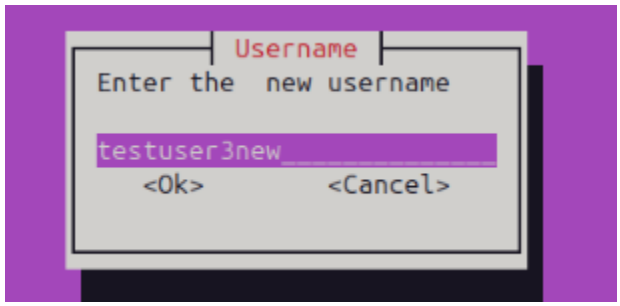

```
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$ sudo ./project.sh
You chose: Add User
testuser3 has been successfully added
```

```bash
if [ "$OPTION" == "Add User" ]
then
        username=$(whiptail --title "Add User" --inputbox "Enter the username" 7 30 3>&1 1>&2 2>&3)
        if [ $? -eq 0 ]
        then
                id $username &> /dev/null
                if [ $? -eq 0 ]
                then
                        echo $username already exists
                else
                        useradd $username
                        echo $username has been successfully added
                fi
        fi
```

3- Modify User



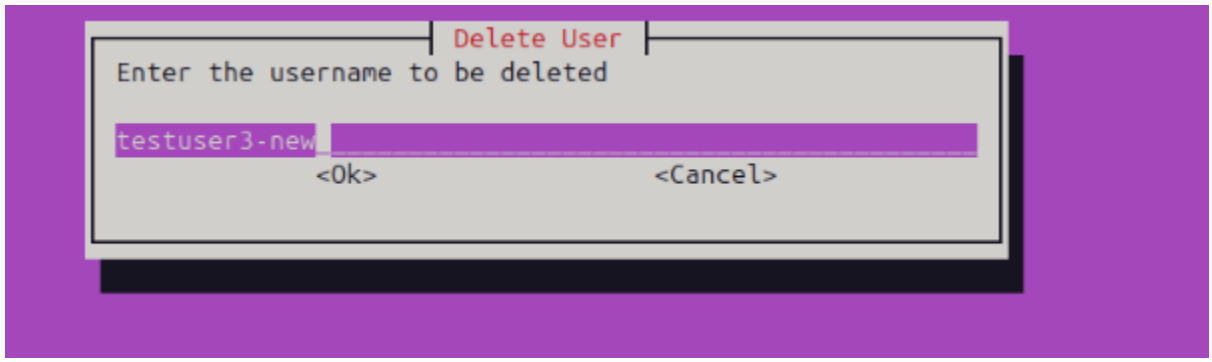





```
elif [ "$OPTION" == "Modify User" ]
then
        username=$(whiptail --title "Modify User" --inputbox "Enter the username" 7 30 3>&1 1>&2 2>&3)
        id "$username" &> /dev/null
        if [ $? -eq 0 ]
        then

                Mod_OPTION=$(whiptail --title "Modify User" --menu "Choose an option" 15 60 4 \
                "Username" "Change the user's name" \
                "UID" "Change the user ID" \
                "Shell" "Change the user's shell" \
                3>&1 1>&2 2>&3)

                case "$Mod_OPTION" in
                        "Username")
                                Required_value=$(whiptail --title "Username" --inputbox "Enter the  new username" 7 3
0 3>&1 1>&2 2>&3)
                                ;;
                        "UID")
                                Required_value=$(whiptail --title "UID" --inputbox "Enter the  new UID" 7 30 3>&1 1>&
2 2>&3)
                                ;;
                        "Shell")
                                Required_value=$(whiptail --title "Shell" --inputbox "Enter the  new shell" 7 30 3>&1
 1>&2 2>&3)
                                ;;
                esac
```

```
            if [ "$Mod_OPTION" == "Username" ]
            then
                    if [ -d "/home/$username" ];
                    then
                    # If it exists, change name and move the folder
                            usermod -l "$Required_value" -d "/home/$Required_value" -m "$username"
                            groupmod -n "$Required_value" "$username"
                    else
                    # If it doesn't exist, just change the name and update the path in /etc/passwd
                            usermod -l "$Required_value" -d "/home/$Required_value" "$username"
                            groupmod -n "$Required_value" "$username"
                    fi
            elif [ "$Mod_OPTION" == "UID" ]
            then
                    usermod -u "$Required_value" "$username"
                    groupmod -g "$Required_value" "$username"
                    chown -R "$Required_value:$Required_value" "/home/$username"

            elif [ "$Mod_OPTION" == "Shell"  ]
            then
                    usermod -s "$Required_value" "$username"
            fi
    else
            echo "${username} does not exist"
    fi
```

4- Delete User:



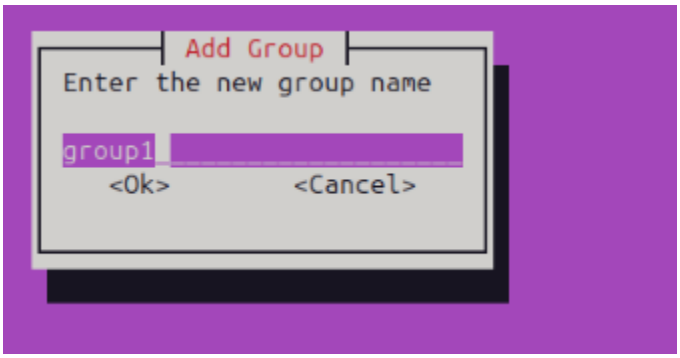

```
elif [ "$OPTION" == "Delete User" ]
then
        username=$(whiptail --title "Delete User" --inputbox "Enter the username to be deleted" 7 60 3>&1 1>&2 2>&3)
        userdel -r "$username"
```

```
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$ sudo ./project.sh
You chose: Delete User
userdel: testuser3-new mail spool (/var/mail/testuser3-new) not found
userdel: testuser3-new home directory (/home/testuser3-new) not found
```

5- List Users:

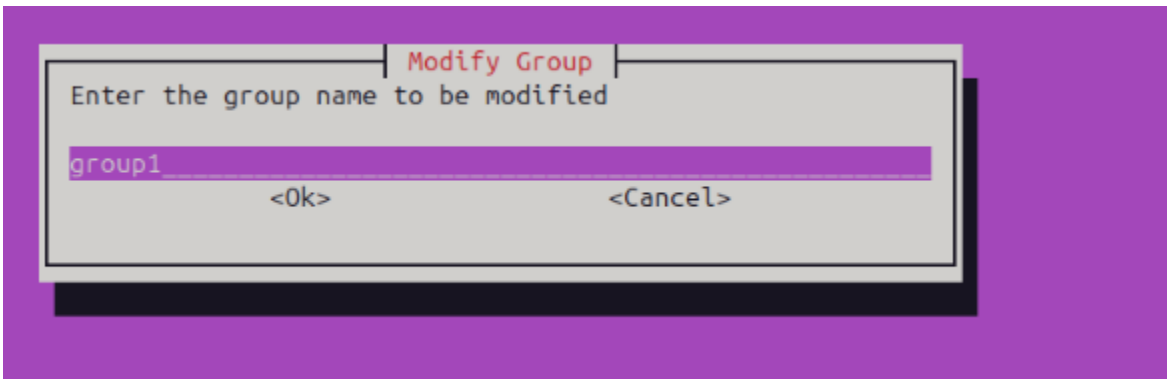
```
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$ sudo ./project.sh
You chose: List Users
root---UID:0
daemon---UID:1
bin---UID:2
sys---UID:3
sync---UID:4
games---UID:5
man---UID:6
lp---UID:7
mail---UID:8
news---UID:9
uucp---UID:10
proxy---UID:13
www-data---UID:33
backup---UID:34
list---UID:38
irc---UID:39
_apt---UID:42
nobody---UID:65534
systemd-network---UID:998
systemd-timesync---UID:996
dhcpcd---UID:100
messagebus---UID:101
syslog---UID:102
systemd-resolve---UID:991
uuidd---UID:103
```

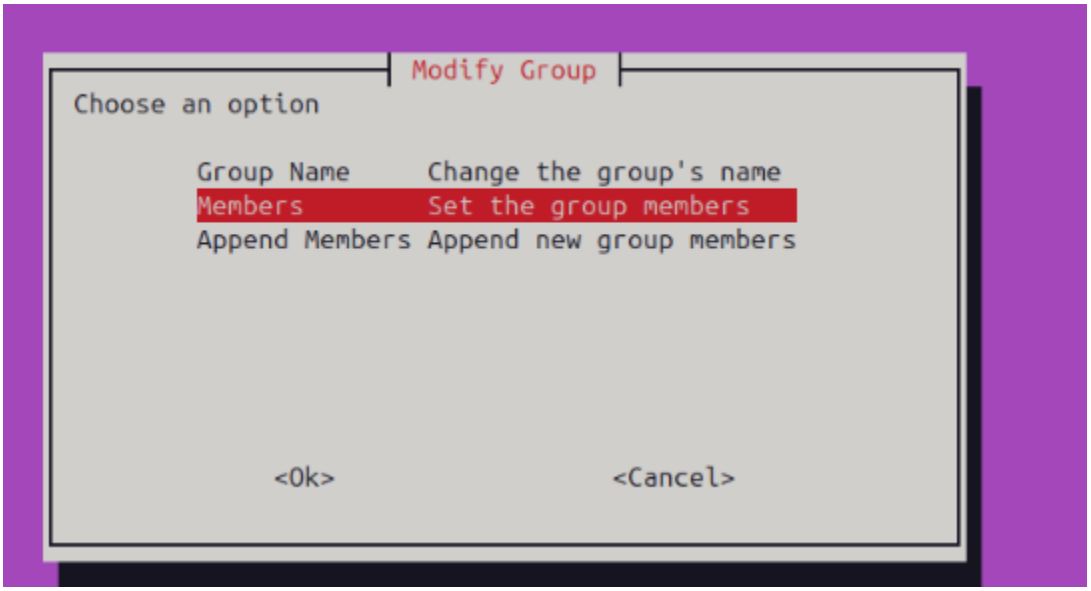
6- Add Group:

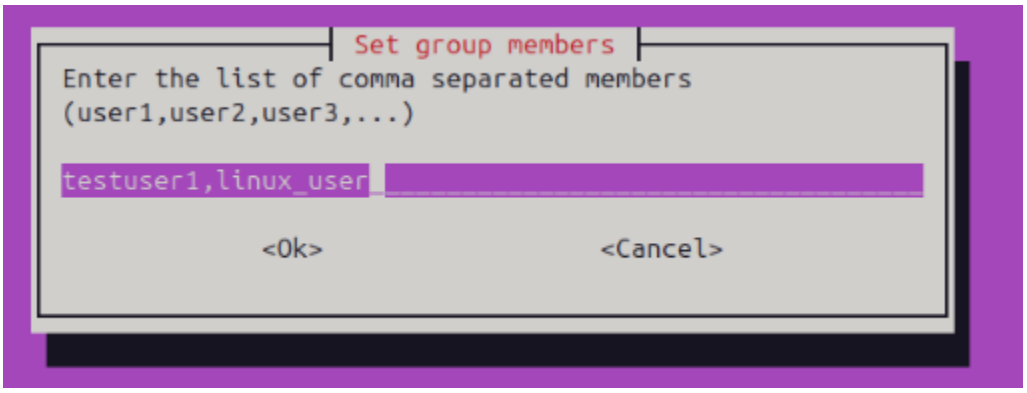



```
elif [ "$OPTION" == "Add Group" ]
then
        group=$(whiptail --title "Add Group" --inputbox "Enter the new group name" 7 30 3>&1 1>&2 2>&3)
        groupadd "$group"
```

7- Modify Group



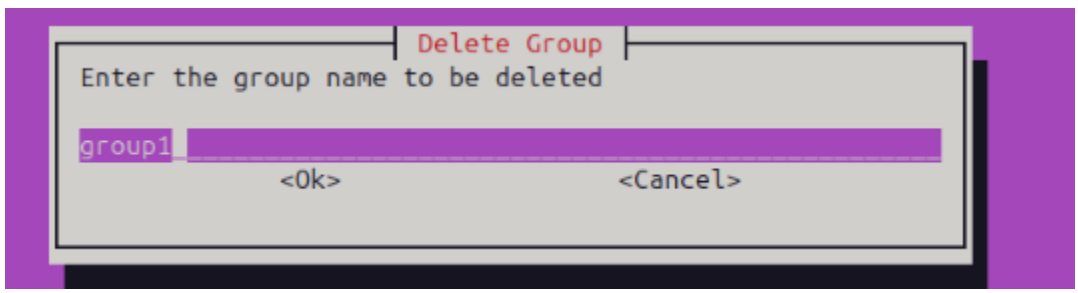





```
group1:x:4325:testuser1,linux_user
```

```
elif [ "$OPTION" == "Modify Group" ]
then
        group=$(whiptail --title "Modify Group" --inputbox "Enter the group name to be modified" 7 60 3>&1 1>&2 2>&3)

        Mod_OPTION=$(whiptail --title "Modify Group" --menu "Choose an option" 15 60 4 \
        "Group Name" "Change the group's name" \
        "Members" "Set the group members" \
        "Append Members" "Append new group members" \
        3>&1 1>&2 2>&3)

        if [ "$Mod_OPTION" == "Group Name" ]
        then
                group_new=$(whiptail --title "Modify Group Name" --inputbox "Enter the new group name" 7 30 3>&1 1>&2
 2>&3)
                groupmod -n "$group_new" "$group"
        elif [ "$Mod_OPTION" == "Members" ]
        then
                members=$(whiptail --title "Set group members" --inputbox "Enter the list of comma separated members
\n(user1,user2,user3,...)" 9 60 3>&1 1>&2 2>&3)
                groupmod -U "$members" "$group"
        elif [ "$Mod_OPTION" == "Append Members"  ]
        then
                members=$(whiptail --title "Append group members" --inputbox "Enter the list of comma separated membe
rs to be appended \n(user1,user2,user3,...)" 9 60 3>&1 1>&2 2>&3)
                groupmod -a -U "$members" "$group"

        fi
                                                                                                  119,4         66%
```

8- Delete Group



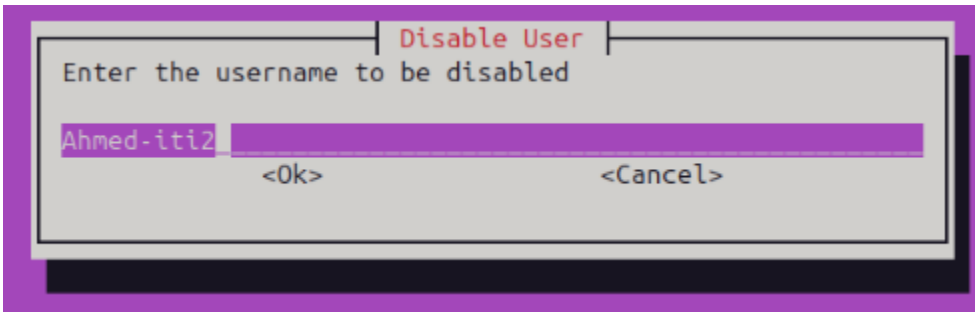

```
elif [ "$OPTION" == "Delete Group" ]
then
        group=$(whiptail --title "Delete Group" --inputbox "Enter the group name to be deleted" 7 60 3>&1 1>&2 2>&3)
        groupdel "$group"
```

9- List Groups

```
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$ sudo ./project.sh
You chose: List Groups
root    GID: 0
daemon  GID: 1
bin     GID: 2
sys     GID: 3
adm     GID: 4
tty     GID: 5
disk    GID: 6
lp      GID: 7
mail    GID: 8
news    GID: 9
uucp    GID: 10
man     GID: 12
proxy   GID: 13
kmem    GID: 15
dialout         GID: 20
fax     GID: 21
voice   GID: 22
cdrom   GID: 24
floppy  GID: 25
tape    GID: 26
sudo    GID: 27
audio   GID: 29
dip     GID: 30
www-data        GID: 33
```

```
elif [ "$OPTION" == "List Groups" ]
then
        awk 'BEGIN{FS=":"}{printf "%s ", $1; printf "   GID: %s \n", $3}' /etc/group
```

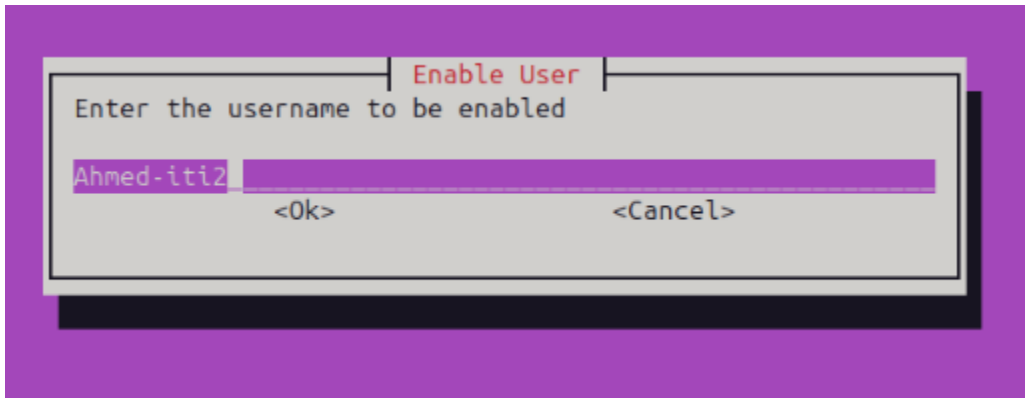
10- Disable User




```
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$ sudo ./project.sh
You chose: Disable User
User Ahmed-iti2 has been locked.
tasneem-adel@tasneem-adel-VMware-Virtual-Platform:~/bash-project$
```

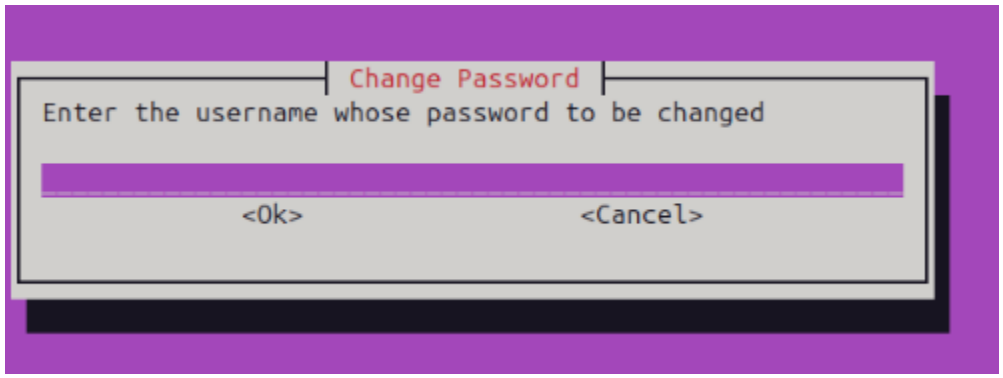
```
elif [ "$OPTION" == "Disable User" ]
then
        username=$(whiptail --title "Disable User" --inputbox "Enter the username to be disabled" 7 60 3>&1 1>&2 2>&3
)
        id $username &> /dev/null
        if [ $? -eq 0 ]
        then
                usermod -L "$username"
                echo "User $username has been locked."
        else
                echo "${username} does not exist"
        fi
```

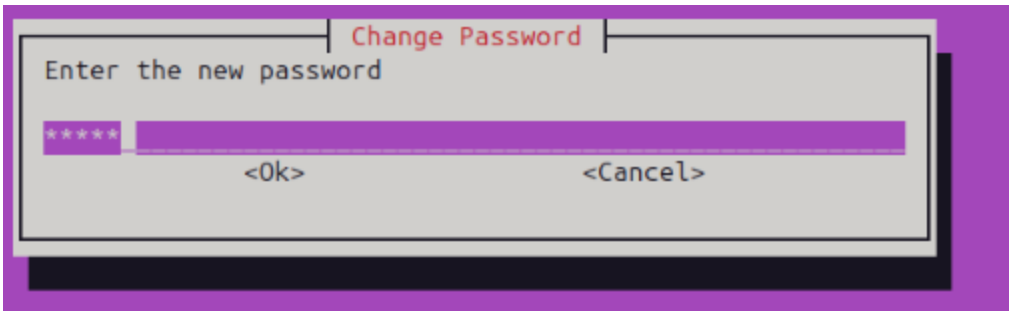
11- Enable User




```
elif [ "$OPTION" == "Enable User" ]
then
        username=$(whiptail --title "Enable User" --inputbox "Enter the username to be enabled" 7 60 3>&1 1>&2 2>&3)
        id $username &> /dev/null
        if [ $? -eq 0 ]
        then
                usermod -U "$username"
                echo "User $username has been unlocked."
        else
                echo "${username} does not exist"
        fi
```

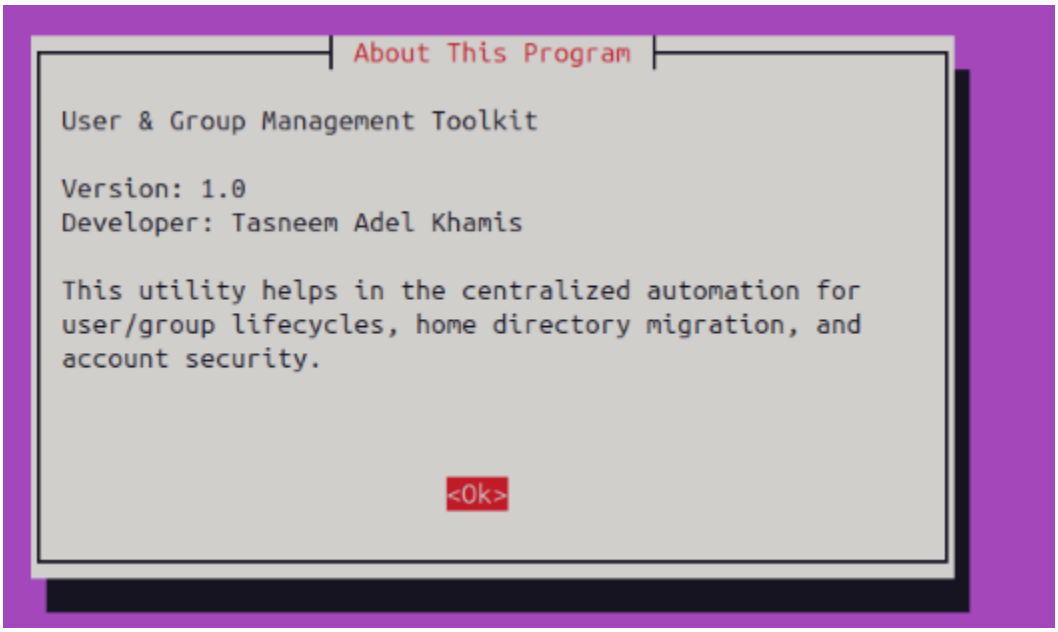
12- Change Password






```
elif [ "$OPTION" == "Change Password" ]
then
        username=$(whiptail --title "Change Password" --inputbox "Enter the username whose password to be changed" 7
60 3>&1 1>&2 2>&3)
        id $username &> /dev/null
        if [ $? -eq 0 ]
        then
                new_pass=$(whiptail --title "Change Password" --passwordbox "Enter the new password" 7 60 3>&1 1>&2 2
>&3)
                echo "$username:$new_pass" | chpasswd
        else
                echo "${username} does not exist"
        fi
```

13- About




```
elif [ "$OPTION" == "About" ]
then
        whiptail --title "About This Program" --msgbox "User & Group Management Toolkit\n\nVersion: 1.0\nDeveloper: T
asneem Adel Khamis\n\nThis utility helps in the centralized automation for user/group lifecycles, home directory migr
ation, and account security." 16 60

fi
"project.sh" 174L, 5919B                                                    174,2          Bot
```